

The minimum bisector energy of a planar point set

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Abstract

The *bisector energy* $E(P)$ of a finite point set $P \subset \mathbb{R}^2$, introduced by Lund, Sheffer, and de Zeeuw, counts ordered quadruples of distinct point pairs sharing a perpendicular bisector. We prove that $E(P) \geq 2n(n-1)$ for every n -point planar set, with equality if and only if every two distinct point pairs have distinct perpendicular bisectors (*bisector-injective*). The bound is tight: for every finite set G in any Euclidean space with no three collinear points, a single linear projection $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ exists such that the image $T(G)$ is bisector-injective, hence attains the floor $E(T(G)) = 2n(n-1)$. The projected set also has zero rotational energy and its set of distinct distances is in bijection with the $\pm$ -difference classes of G . The proof is verified in Lean 4 against standard Mathlib axioms only (propext, Classical.choice, Quot.sound).

1 Introduction

For a finite set $P \subset \mathbb{R}^2$ with $|P| = n$, define the *bisector energy*

$$E(P) = |\{(a, b, c, e) \in P^4 : a \neq b, c \neq e, \beta(a, b) = \beta(c, e)\}|,$$

where $\beta(p, q) = \{x \in \mathbb{R}^2 : |x - p| = |x - q|\}$ is the perpendicular bisector of the pair (p, q) . This quantity was introduced by Lund, Sheffer, and de Zeeuw [4], who proved the upper bound

$$E(P) = O(M(n)^{2/5} n^{12/5+\varepsilon} + M(n) n^2),$$

where $M(n)$ is the maximum number of points of P on any line or circle, and applied it to obtain structural results about point sets with few distinct distances.

The present note addresses the complementary question: *how small can $E(P)$ be?*

Every unordered pair $\{a, b\} \subseteq P$ ($a \neq b$) contributes exactly four ordered quadruples to $E(P)$ by taking both orderings of each pair: (a, b, a, b) , (a, b, b, a) , (b, a, a, b) , (b, a, b, a) . These *trivial quadruples* account for $4\binom{n}{2} = 2n(n-1)$ contributions, so $E(P) \geq 2n(n-1)$ for every P . A nontrivial contribution arises precisely when two distinct unordered pairs share a perpendicular bisector. We say P is *bisector-injective* if no such coincidence occurs, i.e. the map $\{a, b\} \mapsto \beta(a, b)$ is injective on unordered pairs.

The main theorem shows that the floor $2n(n-1)$ is attained.

Theorem 1 (Near Enemy Theorem for Bisector Energy). *Let G be a finite set in a Euclidean space $\mathbb{R}^d$ with no three collinear points, and let $n = |G|$. There exists a linear map $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ such that:*

- (a) T is injective on G ;
- (b) $E(T(G)) = 2n(n-1)$ (bisector-energy floor);
- (c) $E(T(G)) \leq E(P')$ for every n -point planar set P' (absolute minimality);
- (d) $T(G)$ has no three collinear points and no four concyclic points (full general position);
- (e) $E_{\text{rot}}(T(G)) = 0$ (zero rotational energy);
- (f) the set of distinct distances of $T(G)$ is in bijection with the $\pm$ -difference classes of G (distance transport).

The EFPR construction—the lattice-sphere slice $G = \{x \in [-h, h]^d \cap \mathbb{Z}^d : \|x\|^2 = R\}$ of Erdős, Füredi, Pach, and Ruzsa [2]—satisfies the no-three-collinear hypothesis (a line meets a sphere in at most two points), so the theorem applies. The projected EFPR family is the canonical minimizer.

Notation. We write $\langle \cdot, \cdot \rangle$ for the standard inner product on $\mathbb{R}^d$, $\|\cdot\|$ for the induced norm, and $|S|$ for the cardinality of a finite set S . For a linear map $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ we write $T_1, T_2 \in \mathbb{R}^d$ for its two row vectors, so $T(v) = (\langle T_1, v \rangle, \langle T_2, v \rangle)$. The $\pm$ -difference classes of G are the equivalence classes of $G - G \setminus \{0\} := \{a - b : a, b \in G, a \neq b\}$ under $u \sim v$ iff $u = \pm v$.

2 Definitions and the lower bound

Definition 2. A finite set $P \subset \mathbb{R}^2$ is bisector-injective if $\beta(a, b) \neq \beta(c, e)$ for every two distinct unordered pairs $\{a, b\}, \{c, e\} \subseteq P$ with $a \neq b$ and $c \neq e$.

Lemma 3 (Floor and equality). *For every n -point set $P \subset \mathbb{R}^2$:*

$$E(P) \geq 2n(n-1),$$

with equality if and only if P is bisector-injective.

Proof. For each ordered pair (a, b) with $a \neq b$, the quadruples (a, b, a, b) and (a, b, b, a) both lie in the defining set of $E(P)$ since $\beta(a, b) = \beta(b, a)$. As (a, b) ranges over the $n(n-1)$ ordered pairs with $a \neq b$, these $2n(n-1)$ quadruples are pairwise distinct (the first ordered pair (a, b) and the swap bit together uniquely determine the quadruple), so $E(P) \geq 2n(n-1)$.

If P is bisector-injective then every quadruple counted by $E(P)$ has $\beta(a, b) = \beta(c, e)$, hence $\{a, b\} = \{c, e\}$ as unordered pairs, so the quadruple is trivial. Thus $E(P) = 2n(n-1)$.

Conversely, if P is not bisector-injective there exist distinct unordered pairs $\{a, b\} \neq \{c, e\}$ with $\beta(a, b) = \beta(c, e)$. The four ordered quadruples (a, b, c, e) , (a, b, e, c) , (b, a, c, e) , (b, a, e, c) are all counted by $E(P)$ and are nontrivial, giving $E(P) > 2n(n-1)$. $\square$

3 The key criterion: shared bisectors upstairs

The proof of Theorem 1 uses a criterion that translates a shared bisector downstairs into algebraic conditions on the preimages upstairs.

Lemma 4 (Shared-bisector conditions). *Let $p, q, p', q' \in \mathbb{R}^2$. If $\beta(p, q) = \beta(p', q')$ then:*

- (i) $q' - p' = t(q - p)$ for some $t \in \mathbb{R}$ (parallel differences);
- (ii) $\langle p + q - (p' + q'), q - p \rangle = 0$ (midpoint difference orthogonal to direction).

Proof. Two perpendicular bisectors are equal as affine subspaces if and only if they have the same direction and share a point. The direction of $\beta(p, q)$ is the orthogonal complement of $\mathbb{R}(q - p)$; equal directions give (i). Both midpoints $\frac{p+q}{2}$ and $\frac{p'+q'}{2}$ lie on the common bisector, so their difference lies in its direction subspace, giving (ii). $\square$

Remark 5 (Why a single coincidence is not enough). *The orthogonality relation (ii) is only the scalar consequence of a single shared bisector, and on its own it does not force the pairs to coincide: the unit square $\{(0, 0), (1, 0)\}, \{(0, 1), (1, 1)\}$ has no three collinear points and satisfies both (i) and (ii), yet the two pairs are distinct (their common bisector is the line $x = \frac{1}{2}$). Rigidity instead requires the full vector condition $p + q = p' + q'$ (equal midpoints): combined with (i) it places all four points on one line, so a no-three-collinear hypothesis forces the pairs to be equal. The generic-projection argument of the next section supplies exactly this upgrade, by making the orthogonality polynomial vanish for every projection row simultaneously, which promotes the scalar relation to the vector identity.*

4 Generic projections

Definition 6. *A linear map $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ is generic for G if:*

- (inj) $T(a) \neq T(b)$ for all distinct $a, b \in G$;
- (avoid) for every $a, b, c, e \in G$ with $a \neq b$, $c \neq e$, and $\{a, b\} \neq \{c, e\}$: it is not the case that both $T(e) - T(c) = t \cdot (T(b) - T(a))$ for some $t \in \mathbb{R}$ and $\langle T(a) + T(b) - (T(c) + T(e)), T(b) - T(a) \rangle = 0$;
- (sep) for every $a, b, c, e \in G$ with $a - b \neq \pm(c - e)$: $\|T(a - b)\|^2 \neq \|T(c - e)\|^2$.

Lemma 7 (Generic projection implies bisector-injectivity). *If T is generic for G then $T(G)$ is bisector-injective.*

Proof. Suppose $T(a), T(b), T(c), T(e) \in T(G)$ satisfy $\beta(T(a), T(b)) = \beta(T(c), T(e))$ with $T(a) \neq T(b)$ and $T(c) \neq T(e)$. By injectivity of T on G , we have $a \neq b$ and $c \neq e$. By Lemma 4, conditions (i) and (ii) hold downstairs. Linearity of T lifts them: $T(e - c) = tT(b - a)$ and $\langle T(a + b - (c + e)), T(b - a) \rangle = 0$. The avoid condition then forces $\{a, b\} = \{c, e\}$, hence $\{T(a), T(b)\} = \{T(c), T(e)\}$. $\square$

Lemma 8 (Existence of a generic projection). *If $G \subset \mathbb{R}^d$ is a finite set with no three collinear points, a generic projection $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ for G exists.*

Proof. We view the rows $T_1, T_2 \in \mathbb{R}^d$ as $2d$ free real variables. For each off-pair quadruple (a, b, c, e) (with $a \neq b$, $c \neq e$, $\{a, b\} \neq \{c, e\}$), define two polynomials in (T_1, T_2) :

$$\begin{aligned} D(a, b, c, e) &= \langle T_1, b - a \rangle \langle T_2, e - c \rangle - \langle T_2, b - a \rangle \langle T_1, e - c \rangle, \\ H(a, b, c, e) &= \langle T_1, a + b - (c + e) \rangle \langle T_1, b - a \rangle + \langle T_2, a + b - (c + e) \rangle \langle T_2, b - a \rangle. \end{aligned}$$

Condition (avoid) fails for this quadruple exactly at those T where both $D = 0$ and $H = 0$, so it suffices to show $D^2 + H^2$ is not the zero polynomial; its real zero-set then has measure zero. We claim D and H cannot both vanish identically. First, $D \equiv 0$ (as a polynomial in T_1, T_2) if and only if $b - a$ and $e - c$ are linearly dependent, i.e. $e - c = t(b - a)$ for some $t \in \mathbb{R}$. Second, writing $m = a + b - (c + e)$ and $u = b - a \neq 0$, we have $H = \sum_k \langle T_k, m \rangle \langle T_k, u \rangle$, whose associated symmetric form is $\frac{1}{2}(mu^\top + um^\top)$; this vanishes identically if and only if $m = 0$, i.e. $a + b = c + e$. If both held, then $e - c = t(b - a)$ and $a + b = c + e$: letting M be the common midpoint, $a = M - v$, $b = M + v$ with $v = \frac{1}{2}(b - a) \neq 0$, and $c = M - w$, $e = M + w$ with $w = \frac{t}{2}(b - a) = tv$; thus a, b, c all lie on the line $M + \mathbb{R}v$. They are distinct (if $t \in \{1, -1\}$ then $\{c, e\} = \{a, b\}$, excluded), so G would contain three collinear points, contradiction. Hence $D^2 + H^2$ is a nonzero polynomial in $2d$ variables. Similarly, for each pair of distinct $\pm$ -difference classes $[u] \neq [v]$ in $G - G \setminus \{0\}$, the polynomial $\sum_k \langle T_k, u - v \rangle \langle T_k, u + v \rangle$ (encoding condition (sep)) is nonzero: since $u \neq \pm v$, both $u - v \neq 0$ and $u + v \neq 0$, so neither linear factor vanishes identically, and their inner product $\|Tu\|^2 - \|Tv\|^2$ is a nonzero degree-2 polynomial. Likewise the injectivity condition (inj) cuts out a measure-zero set.

The set of $(T_1, T_2) \in \mathbb{R}^{2d}$ satisfying all three conditions is the complement of finitely many measure-zero algebraic sets, hence nonempty (in fact dense): a finite union of proper zero-sets of nonzero polynomials on $\mathbb{R}^{2d}$ has Lebesgue measure zero and dense complement. $\square$

5 Proof of Theorem 1

Proof of Theorem 1. Parts (a)–(c) combine Lemmas 3, 7, and 8: take T generic for G , which exists by Lemma 8. Then T is injective on G (a), $T(G)$ is bisector-injective so $E(T(G)) = 2n(n - 1)$ by Lemma 3 (b), and $2n(n - 1)$ is the global minimum by the lower bound of Lemma 3 (c).

For part (d), one adds two more avoidance conditions to the definition of generic: no collinear triple and no concyclic quadruple in $T(G)$. For collinear triples: $T(a), T(b), T(c)$ are collinear iff $D_{abc}(T) = \langle T_1, b - a \rangle \langle T_2, c - a \rangle - \langle T_2, b - a \rangle \langle T_1, c - a \rangle = 0$. Since a, b, c are not collinear in G , the vectors $b - a$ and $c - a$ are linearly independent, so D_{abc} is a nonzero polynomial. For concyclic quadruples: the 4×4 circle determinant is likewise a nonzero polynomial in (T_1, T_2) , provided no three of the four upstairs points are collinear—a condition guaranteed by the no-three-collinear hypothesis on G . The same measure-zero argument applies.

For part (e), write $E_{\text{rot}}(T(G))$ as the count of ordered quadruples $(p, q, r, s) \in T(G)^4$ with $p \neq q, r \neq s, |p-q| = |r-s|$, and $p-q \notin \{r-s, -(r-s)\}$. The condition $|p-q| = |r-s|$ is equivalent to $\|T(a-b)\| = \|T(c-e)\|$, which expands as $\|T(a-b)\|^2 - \|T(c-e)\|^2 = \sum_k \langle T_k, (a-b) - (c-e) \rangle \langle T_k, (a-b) + (c-e) \rangle = 0$. Condition (sep) forces $a-b = \pm(c-e)$, i.e. $p-q = \pm(r-s)$, so the proper-rotation channel is empty and $E_{\text{rot}}(T(G)) = 0$.

For part (f), the same separation argument shows that the map $T(G) - T(G) \setminus \{0\} \rightarrow G - G \setminus \{0\}$ preserves $\pm$ -classes, giving a bijection between the sets of distinct distances. $\square$

6 Corollaries

Corollary 9 (Sphere-slice form). *Let G be a finite subset of a sphere in $\mathbb{R}^d$, $|G| = n$. Then a linear map $T: \mathbb{R}^d \rightarrow \mathbb{R}^2$ exists satisfying (a)–(f) of Theorem 1.*

Proof. A line meets a sphere in at most two points, so G has no three collinear points. Apply Theorem 1. $\square$

Corollary 10 (EFPR minimizer). *The EFPR lattice-sphere-slice family $G = \{x \in [-h, h]^d \cap \mathbb{Z}^d : \|x\|^2 = R\}$ [2] admits a planar projection that attains the bisector-energy floor. In particular, the minimum bisector energy of an n -point planar set is exactly $2n(n-1)$.*

Proof. Apply Corollary 9 to the EFPR lattice points on a sphere. $\square$

Remark 11 (Relation to LSdZ). *Lund, Sheffer, and de Zeeuw [4] remark in passing that $E(P) = \Omega(n^2)$, but do not extract the constant $2n(n-1)$, state the floor, or characterize equality. The minimization direction (Lemma 3), the equality condition (bisector-injectivity), and the identification of the EFPR projection as a minimizer (Corollary 9) appear to be new.*

Remark 12 (Rotation energy). *Part (e) says the near-enemy projection has zero rotational energy in the sense of the Elekes–Sharir / Guth–Katz framework [1, 3]: every congruent quadruple (a, b, c, e) in $T(G)$ is realized by a translation ($a-b = c-e$) or half-turn ($a-b = -(c-e)$), never by a proper rotation. This is the strongest possible statement: a set with any two congruent pairs at a proper-rotation angle apart contributes to the rotation channel.*

7 Verification

All results in this paper are verified in Lean 4 using Mathlib. The Lean source is `NearEnemyTheorem.lean` in the `NearEnemy` namespace of the `lean-formalizations` repository. The headline Lean theorem corresponding to Theorem 1 is

```
nearEnemy_noThreeCollinear_exists_bisectorEnergy_minimal
  .image_generalPosition_distanceTransport.
```

Running `#print axioms` on this theorem confirms that the proof depends only on `propext`, `Classical.choice`, and `Quot.sound` — the standard Lean/Mathlib axioms. No `sorry` appears anywhere in the proof chain.

References

- [1] G. Elekes and M. Sharir, *Incidences in three dimensions and distinct distances in the plane*, Combin. Probab. Comput. **20** (2011), no. 4, 571–608.
- [2] P. Erdős, Z. Füredi, J. Pach, and I. Z. Ruzsa, *The grid revisited*, Discrete Math. **111** (1993), no. 1–3, 189–196.
- [3] L. Guth and N. H. Katz, *On the Erdős distinct distances problem in the plane*, Ann. of Math. (2) **181** (2015), no. 1, 155–190.
- [4] B. Lund, A. Sheffer, and F. de Zeeuw, *Bisector energy and few distinct distances*, Discrete Comput. Geom. **56** (2016), no. 2, 337–356. arXiv:1411.6868.